Project Plan

QuietSpace

Find Your Focus Anywhere

Industry Partner	N/A
Primary Instructor	Anjana Shah
Team Member	Baris Isci
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1. Executive Summary

QuietSpace is a mobile application designed to help students, remote workers, and freelancers discover reliable, quiet places to study or work outside their homes. The app provides a community-driven platform where users can find, rate, and share productive environments with real-time updates on noise levels, Wi-Fi speed, seating availability, and outlet access.

Objective	Develop a fully functional mobile application that reduces time wasted searching for productive workspaces by providing crowd-sourced, real-time location data with community verification.
Corporate Goals Addressed	Reduce productivity loss for students and remote workers. Build a sustainable community-driven platform. Create a scalable solution that can expand to university partnerships and AI-powered features. Address the market gap where 73% of students struggle to find quiet study spaces and remote workers waste 30-45 minutes daily searching for workspaces
Planned Start Date	Monday, September 15, 2025
Planned End Date	Friday, March 27, 2026

2. Project Approvers, Reviews and Distribution List

Project Role	Name	E-mail	Date
Project Lead	Daniil Orlov	daniil.orlov@georgebrown.ca	October 7
Technical Lead	Ibrahim Yondem	ibrahim.yondem@georgebrown.ca	October 7
Developer	Baris Isci	baris.isci@georgebrown.ca	October 7
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Primary Instructor	Anjana Shah		October 7

3. Scope

In Scope	Out of Scope
Cross-Platform Mobile Development: Native Android app with GPS integration and push notifications (iOS in future release)	Desktop web application
Interactive Mapping System: User-submitted locations with stylized pins, clusters, and real-time updates	Integration with university facility management systems
Comprehensive Check-In System: Real-time reporting on noise level, crowding, WiFi quality using simple sliders and toggles	AI-powered space availability predictions
Location Management: Add/edit locations, favorites list, verification system	QuietSpace Certified partner locations program
User Authentication: Secure sign-in via Google, Apple, or email with customizable privacy settings	Advanced StudyBuddy social platform features
Gamification: Progressive badge system, contribution points, seasonal challenges	Integration with learning management systems (Canvas, Blackboard)

Smart Filtering & Sorting: By quietest, highest-rated, recently updated, study mode compatibility	Virtual study rooms with shared timers
Built-in Pomodoro Timer: Customizable presets integrated with location check-ins	Business partnership revenue models
Optional Social Features: Friend visibility (opt-in), location sharing with study partners	
Trust System: "Last verified X hours ago by Y users" indicators and community ratings	

4. Deliverables

Deliverable	Description
Fully Functional Mobile Application (Android)	Complete mobile app with interactive map, location submission, check-in system, favorites management, filtering/sorting, and user authentication. Native Android application optimized for mobile devices with GPS and location services integration.
Backend Infrastructure	Robust database supporting user-reported data collection (Wi-Fi, outlets, noise, busyness levels), real-time synchronization, user authentication system, and sorting/filtering algorithms. RESTful API for mobile app communication.
Gamification System	Badge system for user engagement, contribution points tracking, check-in rewards, and progress visualization to encourage active participation.
Productivity Features	Built-in Pomodoro timer with customizable work intervals and breaks, integrated with location check-ins for focused productivity sessions.

User Documentation	Comprehensive user guide and tutorial on how to use the QuietSpace mobile application, including all features: map navigation, location submission, check-ins, filtering, Pomodoro timer, and social features.
Technical Documentation	System architecture documentation, API documentation, database schema, deployment guide, and maintenance procedures for future development and scaling.
Project Documentation	Complete set of project documents including Project Vision Document, Project Summary, Test Plans, Sprint Reports, and Final Project Report.

5. Assumptions

This project makes the following assumptions.

- Users will actively participate in community driven content creation and location updates
- Mobile devices (smartphones) will be the primary access method for the application
- Target users have reliable internet connectivity for real-time updates and map functionality
- Location owners (cafés, libraries, coworking spaces) will be receptive to increased foot traffic from app users
- Users will opt-in to location-based services and social features with proper consent
- Critical mass of users can be achieved in target geographic areas (starting with campus locations)

- Educational institutions will be supportive of student productivity tools and may provide partnership opportunities
- Privacy regulations (GDPR, local data protection laws) will allow for location-based social features with proper user consent
- User behavior patterns will provide sufficient data for personalized recommendations
- Team members have access to necessary development tools and resources (IDEs, testing devices, cloud services)
- The development team has sufficient technical expertise in mobile development, backend systems, and database management
- Android development will be prioritized; iOS development is deferred to future releases
- Free tier cloud services and APIs will be sufficient for initial development and testing phases

6. Dependencies

The following are the internal and external dependencies that will have to be acknowledged and addressed:

Internal Dependencies:

- Team member availability and commitment throughout the 6-month project duration
- Access to development environment and necessary software licenses
- Coordination between frontend (mobile) and backend development teams
- Completion of database design before backend API development can begin
- User authentication system must be completed before implementing social features

- Map integration must be functional before location submission features can be fully tested
- Badge system depends on check-in functionality being operational

External Dependencies:

- Third-party mapping API (Google Maps API or Mapbox) availability and rate limits
- Cloud hosting service (AWS, Google Cloud, or Azure) availability and costs
- Authentication services (Firebase Auth, Google Sign-In, Apple Sign-In) functionality
- Push notification service (Firebase Cloud Messaging) reliability
- Internet connectivity for real-time data synchronization
- Mobile OS updates (Android) may require app updates for compatibility
- App store approval process and guidelines (Google Play Store)
- Beta testing user recruitment and participation
- Academic calendar and semester schedule for project milestones

Data Dependencies:

- Initial seed data for locations to demonstrate app functionality
- User-generated content for meaningful testing and validation
- Sufficient check-in data to validate gamification and verification systems

7. Risk Management

Potential Risk	Severity (H/M/L)	Likelihood (H/M/L)	Management Strategy
Low initial user adoption / insufficient critical mass	H	M	Start with focused geographic area (campus locations). Implement incentive program for early adopters. Partner with student organizations for promotion. Seed initial location data to demonstrate value.
Poor data quality from user submissions	H	M	Implement verification system with "last updated" timestamps. Require minimum badge level to add new locations. Use community voting to flag inaccurate data. Monitor and moderate submissions actively.

Team member unavailability or attrition	H	L	Document all code and decisions thoroughly. Use version control (Git) consistently. Cross-train team members on multiple components. Establish clear communication protocols and backup plans.
Technical complexity exceeds team capability	M	M	Break complex features into smaller, manageable tasks. Allocate extra time for learning and experimentation. Seek instructor/mentor guidance early. Use proven frameworks and libraries rather than building from scratch.

Third-party API limitations or costs	M	M	Research API pricing tiers and rate limits during planning phase. Implement caching to reduce API calls. Have backup API providers identified. Design system to be API-agnostic where possible.
Privacy and security vulnerabilities	H	L	Follow security best practices from the start. Use established authentication providers. Conduct security review before launch. Implement proper data encryption. Comply with privacy regulations.
Scope creep / feature bloat	M	H	Strictly prioritize MVP features. Use project plan as scope control document. Defer non-essential features to future releases. Regular sprint reviews to assess progress vs. timeline.

Backend scalability issues	M	L	Design database schema for scalability from the start. Use cloud services that can scale. Implement load testing before launch. Plan for horizontal scaling if needed.
Integration challenges between components	M	M	Define clear API contracts early. Use consistent data formats. Implement integration testing throughout development. Regular system-wide testing.
Timeline delays due to unexpected technical issues	M	M	Build buffer time into schedule (2-4 weeks). Prioritize core features over enhancements. Have weekly progress reviews to identify delays early. Be prepared to reduce scope if necessary.

Device compatibility issues	L	M	Test on multiple Android devices and OS versions. Use standard Android development practices. Implement graceful degradation for older devices.
Competing apps launched during development	L	M	Focus on unique value proposition (community-driven, real-time updates). Monitor competitor landscape. Emphasize features that differentiate QuietSpace.

8. Communication

Reporting

The following reports will be produced;

Report	Audience	Frequency
Sprint Progress Report	Team members, Primary Instructor	Bi-weekly (every 2 weeks)
Weekly Status Update	Team members	Weekly (every Monday)
Risk and Issue Log	Team members, Primary Instructor	As needed / Weekly review
Milestone Completion Report	Team members, Primary Instructor	At each milestone (6 times)
Monthly Project Summary	Team members, Primary Instructor	Monthly
Final Project Report	Team members, Primary Instructor, Stakeholders	End of project (April 2026)

Meetings

The following meetings/communication will be established;

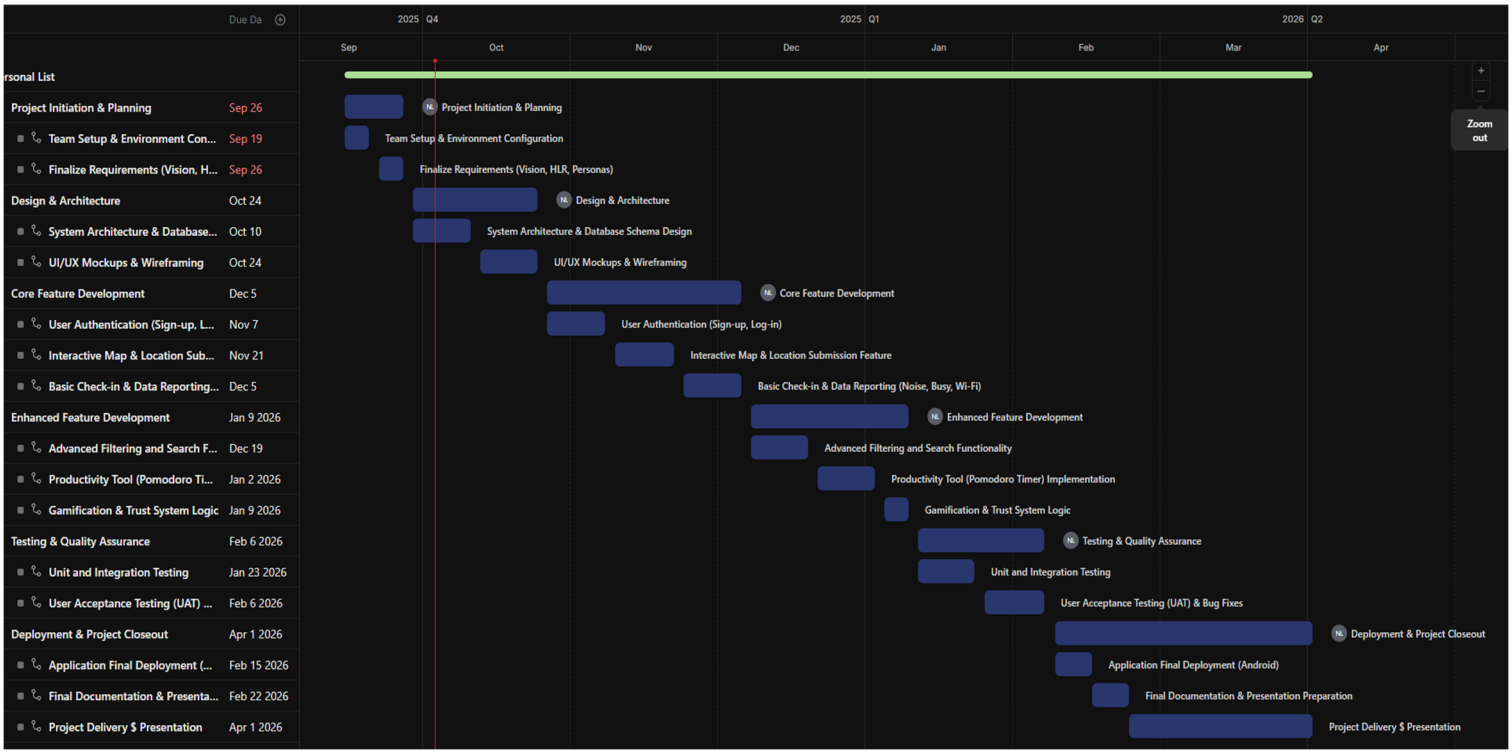
Meeting	Purpose	Attendees	Frequency
Daily Standup	Quick status updates, blockers, daily goals	All team members	Daily (15 minutes)
Sprint Planning	Plan upcoming sprint tasks, estimate effort, assign work	All team members	Beginning of each sprint (bi-weekly)
Sprint Review	Demo completed work, gather feedback	All team members, Primary Instructor (optional)	End of each sprint (bi-weekly)
Sprint	Reflect on process,	All team	End of each

Retrospective	identify improvements	members	sprint (bi-weekly)
Technical Sync	Discuss architecture, integration, technical challenges	Technical team members (Daniil, Baris, Triet)	Twice weekly
Instructor Check-in	Project progress review, guidance, issue escalation	All team members, Primary Instructor	Weekly or bi-weekly
Milestone Review	Major deliverable review and approval	All team members, Primary Instructor	At each major milestone

9. Task Listing (WBS- Work Breakdown Structure)

Reference	Tasks	Duration(Weeks)	Dependency
A	M1: Project Initiation & Planning A.1 Team Setup & Environment Configuration A.2 Finalize Requirements (Vision, HLR, Personas)	2	None
B	M2: Design & Architecture B.1 System Architecture & Database Schema Design B.2 UI/UX Mockups & Wireframing	4	A.2
C	M3: Core Feature Development C.1 User Authentication (Sign-up, Log-in) C.2 Interactive Map & Location Submission Feature C.3 Basic Check-in & Data Reporting (Noise, Busy, Wi-Fi)	6	B.1 B.2
D	M4: Enhanced Feature Development D.1 Advanced Filtering and Search Functionality D.2 Productivity Tool (Pomodoro Timer) Implementation D.3 Gamification & Trust System Logic	5	C.3 C.2 D.1 D.1
E	M5: Testing & Quality Assurance E.1 Unit and Integration Testing E.2 User Acceptance Testing (UAT) & Bug Fixes	4	D.3 E.1
F	M6: Deployment & Project Closeout F.1 Application Final Deployment (Android) F.2 Final Documentation & Presentation Preparation	2	E.2

10. Gantt Chart



11. Milestones

Major Activity or Milestone	Estimated Milestone Target date	Owner/Reviewer Team Members
M1: Project Initiation Complete - Team formed, environment setup, technology stack selected	September 30, 2025	Ibrahim (Owner), All team members (Contributors)
M2: Design Phase Complete - System architecture, database schema, UI/UX mockups, API design finalized	October 31, 2025	Daniil (Owner), Ibrahim, Gozde (Reviewers)
M3: Core Features Complete - Authentication, map integration, location submission, basic check-in functional	December 15, 2025	Baris (Owner), Daniil, Triet (Reviewers)
M4: Enhanced Features Complete - Advanced check-in, filtering, Pomodoro timer, gamification, push notifications implemented	January 31, 2026	Triet (Owner), Baris, Gozde (Reviewers)
M5: Testing Complete - All testing phases finished, bugs resolved, app ready for deployment	February 28, 2026	Gozde (Owner), All team members (Reviewers)
M6: **Project Deployment & Delivery** - Documentation complete, app deployed, final presentation delivered	April 1, 2026	Ibrahim (Owner), All team members (Contributors)

12. RAM – Responsibility Assignment Matrix

Project Team Responsibilities					
Project Name:	QuietSpace				
Project Manager:	Anjana Shah				
Milestone	Ibrahim Yondem (Project Lead)	Daniil Orlov (Technical Lead)	Baris Isci (Developer)	Gozde Baran (Developer)	Nguyen Minh Triet Luu (Developer)
M1: Initiation Complete	R, A	C	I	I	I
M2: Design Phase Complete	C	R, A	I	C	I
M3: Core Features Complete	C	C	R	I	C
M4: Enhanced Features Complete	I	C	C	C	R
M5: Testing Complete	C	I	I	R, A	I
M6: Deployment & Delivery	R, A	C	I	I	I
R: Responsible, A: Accountable, C: Consulted, I: Informed					

13. Approvals

The signatures below indicate their approval of the contents of this document.

Project Role	Name	Signature	Date
Project Lead	Daniil Orlov	Daniil Orlov	October 7
Technical Lead	Ibrahim Yondem	Ibrahim Yondem	October 7
Team Member	Baris Isci	Baris Isci	October 7
Team Member	Gozde Baran	Gozde Baran	October 7
Team Member	Nguyen Minh Triet Luu	Nguyen Minh Triet Luu	October 7
Primary Instructor	Anjana Shah		October 7